

Borked3DS-rpi optimizations since the letters bug (v146 → v399)

Raspberry Pi 5, V3DV / Mesa 26.1.2, RetroPie. This lists the **kept** changes that improve the emulator. New in this edition: v398 and v399 (fixes ported from gvx64's Pi 4 fork) and one install script for the Pi 4 and the Pi 5 (section 10).

Legend

Mark	Meaning
✓	Applies. It was measured, or the code is shared and active in that renderer.
(✓)	The code is shared by both renderers, so it probably applies. It was never measured under OpenGL/GLES.
—	Does not apply (Vulkan-only or OpenGL-only code).

- **Speed gain:** given as % of speed (frames per second), computed from the project's measurements.
- **Cold loading time:** given as % less time.
- All % values are **approximate**. They depend on the game and scene measured.
- Speed gains do not add up. Each one was measured on a different game and version.

1. In-game speed

Version	Improvement	Measured gain	≈ %	Vulkan	OpenGL (GLES)
v149c / v150	Scheduler mode C: no GPU Finish after each software draw (EmuThread was blocked 60-69 % of the time)	Metroid 37-40 % → 60 % speed	+55 % speed	✓	—
TB33	Removed the image copy (CopyImageView) on every draw	Metroid 49.3 % → 88.7 %	+80 % speed	✓	—
TB34	Heavy traces made opt-in (off by default)	88.7 % → ~95 %	+7 % speed	✓	—
v158 / v389	Lighter log filter (*:Trace → *:Warning by default)	+5 speed points	≈ +5 % speed	✓	✓
v160	Environment variables read once instead of on every draw (pica_core, rasterizer, pipeline cache)	EmuThread CPU 68.1 % → 54.5 %	−20 % CPU (no fps gain; the CPU was not the bottleneck then)	✓	—
v148	SAFE_PICA_HW budgets raised (1/6 → 9999/65536): every draw goes through the GPU	Render bottleneck removed	not quantified	✓	—
v388	Robustness disabled on the fragment shader only (VK_EXT_pipeline_robustness): 1278 → 839 instructions, spills 16:27 → 2:2	Luigi's Mansion 2: 54 → 37.8 ms per frame	+43 % speed	✓	—

Version	Improvement	Measured gain	≈ %	Vulkan	OpenGL (GL ES)
v396 / v397	CPU hot path trimmed after a perf profile of Luigi's Mansion 2: identical PICA shader words no longer mark the program dirty; index min/max in NEON; environment lookups cached once, without building a string on each call; descriptor writes batched under strict-compat (as upstream)	Luigi's Mansion 2: 32.5 → 31.3 ms per frame (real time, TB101/TB104); other games unchanged at 16.7 ms	≈ +4 % speed (Luigi)	✓	(✓) partly shared (PICA, min/max)
v391	Reduced lighting for Luigi's Mansion 2 only (title 0004000000076500): 6 of the 8 per-pixel lights are computed. All other games are untouched.	Luigi's Mansion 2: ~40 → 26 ms per frame (census method, see note in section 7); slow pipeline compiles 0.9-3.4 s → up to 1.7 s	+50 % speed (Luigi only)	✓	—
v398	Guest memory hot path (gvx64 34629e9 + 41eca49): the plugin-loader service is looked up once instead of by name on every access to GPU-cached memory, and those accesses use a raw pointer (no shared_ptr copy, atomics or virtual calls)	Smash 16.8 ms (TB108, unchanged: already at 60 fps); measured together with v399 on Luigi, see below	CPU work removed; gain shows only in CPU-bound games	✓	✓
v399	Real 3DS timings for system calls (gvx64 99a0d7e = Azahar #1093): each system call advances the emulated clock by the cycle count measured on real hardware, instead of a fixed +150 ticks in GetSystemTick only. Games that busy-wait for the end of a frame reach it with fewer emulated instructions.	Luigi's Mansion 2 garden, A/B in the same session: 41.9 → 37.5 ms per frame (TB109). Official power supply, v398 + v399: garden 31.3 → 30.4 ms (29.4 ms at the end of the run, TB110)	≈ +12 % speed (A/B); ≈ +3 % vs v397	✓	✓ (emulated kernel, renderer-independent)

2. Freezes, stutter and smoothness

Version	Improvement	Measured gain	≈ %	Vulkan	OpenGL (GL ES)
v151	Extended Dynamic State (EDS) enabled: fewer pipeline variants	Kid Icarus worst compile 12,993 → ~51 ms; poisons 43 → 0	-99.6 % of worst freeze	✓	—
v326 / v329	No blocking wait on BindPipeline , plus fix for is_pending never being reset	8-20 s freezes removed	freezes removed	✓	—
v331 / v342	Asynchronous compile pool, pool-inversion fix, 6 worker threads	Compilation = 0.8 % of the session; no shader waits	not quantified	✓	—
v361	glslang internal optimizer turned off (unbounded exhaustive inlining)	Major freeze removed (12 GiB runaway allocation on EmuThread, frozen console)	major freeze removed	✓	—
v371	Short wait for the first pipeline instead of skipping the draw	Frames with skipped draws 28 → 4; early 2D flash fixed	-86 % damaged frames	✓	—
v372	In-memory and on-disk SPIR-V cache (38 % of conversions were redundant)	Warm: ~1 s 3D freeze removed; frames > 100 ms 8 → 2	-75 % warm stutter	✓	—
v390	Compile threads at background priority (nice 19), 3 threads by default: compiling never steals CPU time from emulation	Kid Icarus cold: 31 → 18 s of compile time, 12 frames > 80 ms; Luigi: frames > 80 ms 117 → 8 during compiles	-42 % cold compile (Kid Icarus)	✓	—

Version	Improvement	Measured gain	≈ %	Vulkan	OpenGL (GLES)
v392	Vulkan shader caches (pipelines + SPIR-V) are kept across rebuilds ; only the SPIR-V part is purged if the GLSL → SPIR-V conversion itself changes	Areas already visited never recompile, even after an emulator update	no repeated cold start	✓	—
v383 / v384	GPU lead bounded, pipeline cache saved during gameplay	Luigi flicker removed; cache stays warm even after a crash	not quantified	✓	—
v398	Real-time audio hang fixed (gvx64 df3bca6): with <i>realtime audio</i> on, a long frame (shader compile) scaled the audio tick without limit and flooded the game with audio interrupts (game at 0 fps, hang). The scale is now capped at 3× and the audio tick can never fire back-to-back. Log line V398_AUDIO_BORNE when the cap is hit.	Hang reproduced and fixed by gvx64 on the Pi 4 (ALBW, FE Awakening, empty shader cache). Not measured on the Pi 5 (option off by default).	hang removed	(✓)	✓

3. Cold shader loading time

Version	Improvement	Measured gain	≈ %	Vulkan	OpenGL (GLES)
v377	Acyclic PICA decompiler (state machines without poison loops)	Sonic poisons 77 → 11 s	-86 % loading	✓	(✓)
v385	Vertex-shader boolean and loop specialization, cap 64	Sonic 62 → 9 s	-85 % loading	✓	(✓)
		Kid Icarus 319 → 19-31 s	-90 to -94 % loading	✓	(✓)
		Zelda 68 → 12 s	-82 % loading	✓	(✓)
		Metroid 12 → 5 s	-58 % loading	✓	(✓)

4. Games made playable or fixed

Version	Improvement	Measured gain	Vulkan	OpenGL (GLES)
v379 / v380	Backward jumps and specialized JMPU jumps in the decompiler	Luigi's Mansion 2: 98.6 s black screen then crash → playable	✓	(✓)
v313 / v335	HIGH_SWITCH / DIRA removed: avoids the software vertex path	Sonic 3D runs on the GPU	✓	—
v395	Seven small fixes ported from Azahar: cull mode value 3 (KeepAll2) no longer crashes the emulator; zero-area draws skipped; invalid vertex-array addresses no longer read; disabled cube texture units get a null cube; malformed geometry shaders (ARM JIT); FillScreen before the screen image exists; nearest-neighbour present sampler kept nearest	Crash and correctness fixes; no speed change (TB99: 5 test games, 0 assertion)	✓	✓ (cull mode, vertex arrays, GS)
v394	Logic op NoOp honoured in Vulkan : under strict-compat the hardware logic op is off and emulated in the shader, but NoOp ("leave the color alone") cannot be emulated there. It now sets the color write mask to 0, keeping depth and stencil writes.	Super Mario 3D Land: shadows drawn as long black spikes → correct shadows (TB97). No speed cost (TB98: Luigi 32.5 ms with and without).	✓	— (GLES was already correct through glColorMask)

Version	Improvement	Measured gain	Vulkan	OpenGL (GLES)
v398	GLES black boxes fixed (gvx64 3fad3e0, ccd7cc2): <code>glCopyImageSubData</code> is now resolved on GLES (<code>GL_EXT/OES_copy_image</code> ; the pointer was null and crashed); <code>GetCurState()</code> returns the real GL state; <code>CopyTextures</code> uses state-managed framebuffers (no more 0x502 and blacklisted colour buffer)	gvx64 on the Pi 4: black box and transparent 3D fixed in Fire Emblem Awakening/Fates, Yo-kai Watch, Dragon Ball Fusions. Not measured on the Pi 5.	—	✓
v398	GL geometry shader used only when the driver exposes <code>GL_EXT_geometry_shader</code> (desktop GL always has it)	Driver-correct GS choice under GLES	—	✓

5. System and settings

Version	Improvement	Measured gain	Vulkan	OpenGL (GLES)
v324	<code>cpu_clock_percentage</code> 60 → 100	Games run at their nominal pace	✓	✓
v376	Mesa 26.1.2 kept (rather than 26.1.6)	Better of the two drivers tested	✓	(✓)
v378	Official 27 W power supply	Pi crashes removed	✓	✓
v389	Optimal settings built into the code. Vulkan and disk cache on by default, fullscreen on. The 17 V3DV variables are set automatically, in Vulkan only. <code>emulators.cfg</code> reduced to 3 lines.	Same speed, simple setup	✓	✓ (Vulkan is the default; GL can still be selected)
v398	Idle texture surfaces evicted (gvx64 4922386): surfaces unused for 3600 frames (~60 s) are freed, at most once every 300 frames and only above 256 cached surfaces. Opt-out: <code>BORKED3DS_V3DV_V398_NO_IDLE_EVICT=1</code> .	gvx64, Pi 4 8 GB: Pokémon Sun peak RAM 8 GB+ → under 3 GB. Pi 5: no speed loss (Smash 16.8 ms, TB108).	✓	✓
v398	One install script for the Pi 4 and the Pi 5, Bookworm and Trixie (see section 10)	—	✓	✓

6. Tried, then removed (no gain or a regression)

Version	Attempt	Why it was removed	Vulkan	OpenGL (GLES)
—	<code>opt_compile_time</code>	Halved compile time. With the non-robust FS it gave 1336 instructions and Luigi at 126 ms.	✓	—
v382	TEXSYNC	Superseded by v383/v384	✓	—
v386	Forced early-Z	52.9 → 52.1 ms, negligible gain	✓	—
v386	SKIP_DARK (skip dark lights)	Slower, and the bottom screen broke	✓	(✓)
—	Looped or grouped lighting	1356 to 1760 instructions vs 839	✓	(✓)
—	<code>glslang -Os</code>	Identical after NIR	✓	—
v351	<code>optimize_spirv_output</code> modes 1 to 3	34 s poisons and GPU hangs; stays at 0	✓	—
TB87	Geometry shader turned off (<code>geometry_shader=false</code>)	No gain: the fragment lighting shader is the cost, not the GS	✓	—

Version	Attempt	Why it was removed	Vulkan	OpenGL (GLES)
TB88	2 compile threads (normal priority)	Smooth, but the compile queue never caught up (most draws skipped); replaced by v390	✓	—
TB91a	Luigi with 4 lights	Barely faster than 6 lights (CPU-bound) and far too dark	✓	—
Azahar lot B	Surface garbage collection driven by GPU ticks (would remove the GPU drain in the surface destructor)	Measured first (TB100b): 298 drains in 144 s of Luigi cost 139 ms in total (~0.1 %). Most of the other lot B commits fix code this fork does not have. Not ported.	✓	—
Azahar #1059 / #2318	Register state tracking and command-list processing rewrites	Profile: PICA register handling is only ~5 % of the time, worth about 1 ms, for a large and risky port. Not ported.	✓	(✓)
TB101	Luigi with the original 8 lights on v396	42.3 ms vs 31.5 ms: the lighting cost is on the GPU, CPU work cannot recover it	✓	—
v393	PICA shadow maps rendered in Vulkan (atomic writes to an r32ui image, Shadow2D/ShadowCube reads)	Did not fix Mario 3D Land, which never uses PICA shadow maps. Luigi's Mansion 2 (flashlight shadow pass every frame) went from 32.5 to 35.4 ms. Kept in the code but off by default since v394; <code>BORKED3DS_V3DV_V393_SHADOWS=1</code> turns it on.	✓	—
gvx64	Not ported: Wait Tree debugger pause (bc226eb); Pi 4 GLES-only paths (MaxPerformance shader mode, FE Awakening light collapse, emergency software fallback and its hotkey), which conflict with this fork and target the Pi 4's GPU	Pi 4 users get them from gvx64's fork through the install script	—	—

7. Final result (v399, Vulkan; 60 fps = 16.7 ms per frame, warm caches, official 27 W power supply)

Game	ms per frame	Cold shader loading
Kid Icarus Uprising	16.6 (full speed)	~18 s of compile time, 0 poisons
Metroid	16.7 (full speed)	< 1 s
Zelda	16.7 (full speed)	12.5 s
Smash Bros	16.8 (full speed, v398, TB108)	7-10 s
Sonic	16.7 (full speed)	~5 s
Luigi's Mansion 2 (6 lights)	garden 30.4 in real time (~55 % speed; v399, TB110); opening scene ~44	1-2 s per new pipeline in unvisited areas
Super Mario 3D Land	shadows correct since v394	—

Luigi's Mansion 2 is at its ceiling on this Pi. A [perf](#) profile (TB102/TB103) shows the emulation thread busy about 79 % of a core, with its time spread over dozens of functions (no single hot spot), while the GPU is loaded by the per-pixel lighting. With the original 8 lights it runs at 42.3 ms per frame (TB101), so v391's 6 lights stay.

Measurement notes.

- Census vs real time (TB98).** The 26 ms figures for Luigi come from the census [frame_us](#), which samples only one frame in sixty. Measured from real elapsed time over 60 frames ([V383_LAG](#) timestamps), the same garden scene ran at 32.5 ms per frame (31.3 ms in v397, 30.4 ms in v399). Only the real-time figure matches what

the player sees; Luigi measurements use it. The relative gains above (v388, v391) were each measured with a single method and remain valid.

- **Measure scene by scene (TB109/TB110).** A Luigi run has an opening scene at ~44 ms for about a minute, a cut-scene at 16.7 ms, then the garden at ~29-31 ms. The median of a whole run mixes them and means nothing; each scene is compared on its own.
- **Battery.** Figures taken with the Pi on a battery are not compared with the references, which were all taken on the official 27 W supply. An A/B comparison made in the same session (same battery, same binary) remains valid; that is how v399 was measured.

Known defect: small flashes on Smash's bottom screen. They existed before v388.

8. Validation at stock clocks (TB92: 2.4 GHz CPU, 960 MHz GPU)

Most Pi 5 owners do not overclock. The reference Pi runs `arm_freq=3000`, `v3d_freq=1000` and `over_voltage_delta=50000`; these were disabled for this test (`arm_boost=1`, a stock Raspberry Pi OS setting, was kept).

Game	3.0 GHz / 1000 MHz	2.4 GHz / 960 MHz	Reading
Luigi's Mansion 2 (6 lights)	26.0 ms (census; ~31-32 ms in real time)	33.8 ms (~49 % speed)	CPU and GPU both slower; GPU busy 93 %
Kid Icarus	16.6 ms	16.6 ms	Full speed, no loss
Smash Bros	16.8 ms	median 16.6 ms , mean 19.1 ms	Full speed most of the time; ~10 % of frames above 22 ms

No throttling (`throttled=0x0`), 49-51 °C, official 27 W power supply. Overclocking gives Luigi's Mansion 2 about +30 % speed; the other games do not need it. (Measured before v398/v399.)

9. Cold run and warm run

A **cold run** is the first time a game (or a new area of a game) is played after installation. The 3DS shaders must first be translated and compiled for the Pi's GPU. This happens in the background, so the game keeps running, but while a shader is not ready you may see minor glitches: an object appearing a moment late, a brief flash, a short stutter or a frame that looks frozen for an instant. In Luigi's Mansion 2, each unvisited room can take a few seconds to settle.

A **warm run** is every run after that. The compiled shaders are saved on disk and loaded instantly, so the same scenes play smoothly from the start. Since v392 these caches are also kept when the emulator is rebuilt or updated, so the warm-up only happens once per game area.

In short: the first minutes of a new game may look a little rough; play on, and it settles.

10. Installation on the Pi 4 and the Pi 5 (`borked3ds.sh`, v398)

One RetroPie scriptmodule now installs the right emulator for the board it runs on.

	Raspberry Pi 5 (Pi 500, CM5)	Raspberry Pi 4 (Pi 400, CM4)
Repository	DTEAM-1/Borked3DS-rpi (this fork)	gvx64/Borked3DS-rpi
Starting renderer	Vulkan (code default)	OpenGL with "Use OpenGL ES" ticked (<code>graphics_api=1</code> , <code>use_gles=true</code> , written once in <code>qt-config.ini</code>)

	Raspberry Pi 5 (Pi 500, CM5)	Raspberry Pi 4 (Pi 400, CM4)
Compiler	Clang, <code>-march=armv8.2-a+crc+crypto -O3</code>	System compiler, RetroPie flags, as RetroPie-Extra builds it (no <code>+crypto</code> : the Pi 4 has no ARMv8 crypto extension)
Post-install check	Every fix marker searched in the binary; Vulkan shader cache kept	—

- **Board detection:** `/proc/device-tree/model`, then RetroPie's platform flags. Can be forced with `BORKED3DS_TARGET=pi4` or `pi5`; the repository with `BORKED3DS_REPO=<url>`. 64-bit (aarch64) only.
- **Debian Trixie (13):** system CMake 3.31 and Clang, as before.
- **Debian Bookworm (12):** the tree needs CMake 3.26 or newer and Bookworm ships 3.25, so CMake 3.31.6 is downloaded into RetroPie-Setup's `tmp` folder (nothing installed system-wide). The Pi 5 build uses `clang-16` from Bookworm's repository (the default Clang 14 is too old for this C++23 tree). Bookworm's Mesa 24.x has not been tested with the Vulkan path; a warning is printed.
- **Pi 4 settings:** only the two renderer keys are written, and a key the user already set by hand is left alone. The gamepad and every other setting are never touched.
- **Pi 4 with less than 6 GB of RAM:** compiled on 2 cores to avoid running out of memory.
- **Missing dependencies added:** `robin-map-dev` (required by dynarmic; builds only worked on machines that already had it), `qt6-multimedia-dev`, `qt6-base-private-dev`, `libxcb-cursor-dev` and others; Qt is always taken from the system (`USE_SYSTEM_QT=ON`).

OpenGL (GLES) note

All measurements in this document were taken in Vulkan on the Pi 5. v390 to v394 and the v397 descriptor change are Vulkan-only. v395, v396, v398 (memory path, audio fix, surface eviction) and v399 (system-call timings) are shared code and apply under OpenGL as well. The v398 GLES fixes (copy image, [CopyTextures](#), geometry-shader check) concern OpenGL only and were validated by `gvx64` on the Pi 4. Mario 3D Land shadows were already correct under GLES. The GL driver compiles GLSL itself, so the SPIR-V cache, robustness, EDS and the compile pool do not concern it.

Credits

v398 and v399 port work by [gvx64](#) (`gvx64/Borked3DS-rpi`, the Pi 4 fork): [34629e9](#), [41eca49](#), [df3bca6](#), [4922386](#), [3fad3e0](#), [ccd7cc2](#), [3c02bf2](#) (partly) and [99a0d7e](#) (itself a backport of Azahar #1093, cycle counts profiled on real hardware by PabloMK7).